

SE Lab 02

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Class: 'AIML-D'

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Tables

Epic 1: Subscriber Health Profile & Metric Tracking

Description: Manage patient metabolic constraints, track daily physiological metrics, and enforce strict role-based data encryption for clinical privacy.

User Story	As a	I want to	So that
Story 1.1	Subscriber	Input and update my metabolic profile, daily calorie goal, and allergen exclusions	Input and update my metabolic profile, daily calorie goal, and allergen exclusions
Story 1.2	Subscriber	Log daily blood glucose levels, body weight, and blood pressure readings	My clinical care team can track my vitals and adjust dietary recommendations.
Story 1.3	Sys Admin	Enforce AES-256 encryption at rest, TLS 1.3 in transit, and RoleBased Access Control (RBAC)	Sensitive clinical records and metabolic parameters are protected from unauthorized access.

Epic 2: Automated Meal Plan Generation & Allergen Validation

Description: Programmatically generate balanced weekly menus that strictly respect metabolic constraints, dietary thresholds, and performance targets.

User Story	As a	I want to	So that
Story 2.1	Sys Engine	Generate a personalized 7-day meal plan calculated within 5% of the subscriber's calorie goal	Meal recommendations meet nutritional targets and clinical caloric requirements.
Story 2.2	Sys Engine	Execute automated allergen checks against all suggested menu ingredients before publishing a draft	Any dish containing the subscriber's excluded allergens is filtered out prior to review.
Story 2.3	Platform Architect	Optimize menu generation throughput to sustain 2.5 seconds execution under peak load	The subscription engine handles large batches of automated renewals without latency bottlenecks.

Epic 3: Clinical Review & Dietitian Approval Workflow

Description: Provide certified dietitians with clinical oversight tools to audit, modify, and sign off on generated meal plans.

User Story	As a	I want to	So that
Story 3.1	Dietitian	Inspect draft weekly plans alongside patient macro-nutrient breakdowns, diabetic thresholds, and portion limits	I can verify clinical safety and suitability prior to kitchen preparation.
Story 3.2	Dietitian	Override specific meal items with eligible allergen-free replacements and clinical notes	Unsuitable items (e.g., elevated sodium) can be corrected while automatically recalculating weekly macro totals.
Story 3.3	Dietitian	Formally sign off on a verified meal plan to transition its state to "Approved"	Kitchen dispatch is scheduled and audit logs record my review credentials.

Epic 4: Delivery Schedule Management

Description: Allow subscribers to pause, skip, or reschedule planned meal deliveries while enforcing kitchen dispatch deadlines.

User Story	As a	I want to	So that
Story 4.1	Subscriber	Pause, skip, or reschedule an upcoming daily meal delivery from my account portal	Deliveries match my personal travel schedules and prevent food wastage.
Story 4.2	System Engg	Enforce a strict 12-hour pre-dispatch deadline for all schedule modifications	Kitchen preparation and logistical dispatch operations are not disrupted by late cancellations.

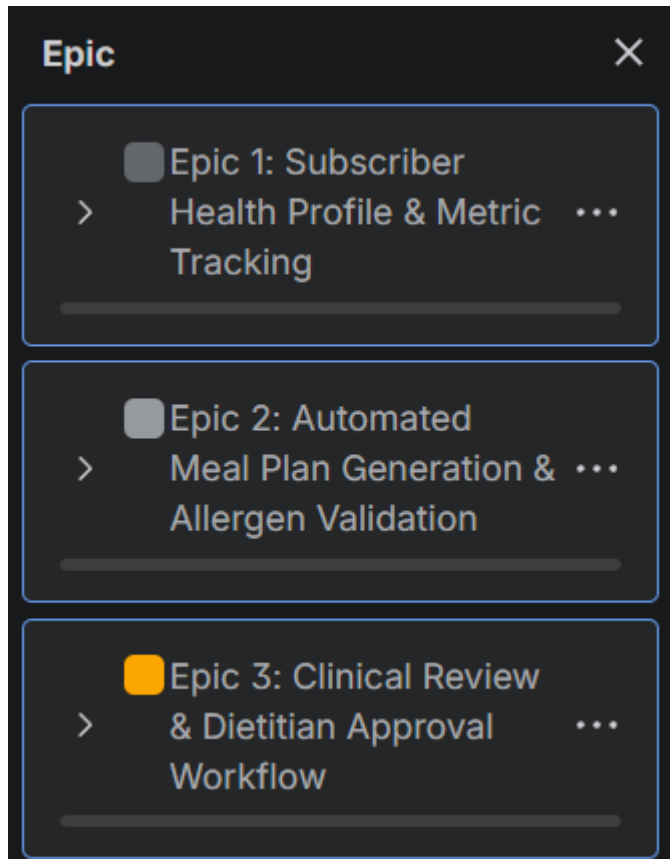
Epic 5: Order Fulfillment & Delivery Tracking

Description: Track daily preparation and fulfillment stages, notifying subscribers in real time across status changes.

User Story	As a	I want to	So that
Story 5.1	Subscriber	View the real-time operational status of my daily meal package	I can anticipate when my food is prepared, dispatched, and delivered.
Story 5.2	Subscriber	Receive instant push/SMS alerts whenever my meal fulfillment status changes	I can coordinate receipt of perishable dietary packages promptly.

Screenshots

Epics and stories created:



Epic

Tracking

Epic 2: Automated Meal Plan Generation & Allergen Validation

Epic 3: Clinical Review & Dietitian Approval Workflow

Epic 4: Delivery Schedule Management

Epic 5: Order Fulfillment & Delivery Tracking

PMDSM Sprint 2

Add dates

(9 work items)

3400

☒	PMDSM-6	Story 1.1: Metabolic Profile and Allergen Configuration	Epic 1: Subscriber ...	To Do	3		
☒	PMDSM-7	Story 1.2: Daily Health Metric Logging	Epic 1: Subscriber ...	To Do	3		
☒	PMDSM-8	Story 1.3: Health Data Encryption & Access Control	Epic 1: Subscriber ...	To Do	5		
☒	PMDSM-9	Story 2.1: Algorithmic Weekly Menu Generation	Epic 2: Automated ...	To Do	5		
☒	PMDSM-10	Story 2.2: Mandatory Allergen Constraint Validation	Epic 2: Automated ...	To Do	3		
☒	PMDSM-11	Story 2.3: High-Concurrency Plan Generation Performance	Epic 2: Automated ...	To Do	5		
☒	PMDSM-12	Story 3.1: Clinical Dashboard Menu Review	Epic 3: Clinical Rev...	To Do	3		
☒	PMDSM-13	Story 3.2: Clinical Item Override & Real-Time Macro Recalculation	Epic 3: Clinical Rev...	To Do	5		
☒	PMDSM-14	Story 3.3: Meal Plan Sign-off & Audit Logging	Epic 3: Clinical Rev...	To Do	2		

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Stories:

1) Story 1.1:

PMDSM-1 / ☒ PMDSM-6

1

Story 1.1: Metabolic Profile and Allergen Configuration

Done ▾

✓ Done

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✦ Improve Story

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▽ Description

◦ As a: Subscriber

◦ I want to: Input and update my metabolic profile, daily calorie goal, and allergen exclusions

◦ So that: The subscription engine can tailor weekly menus strictly around my health constraints.

2) Story 1.2:

PMDSM-1 / ☒ PMDSM-7

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Story 1.2: Daily Health Metric Logging

Done ▾

✓ Done

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✦ Improve Story

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▽ Description

As a: Subscriber

◦ I want to: Log daily blood glucose levels, body weight, and blood pressure readings

◦ So that: My clinical care team can track my vitals and adjust dietary recommendations.

3) Story 1.3:

 PMDSM-1 /  PMDSM-8

 1  

Story 1.3: Health Data Encryption & Access Control

Done ▾

✓ Done



✦ Improve Story

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▽ **Description**

- As a: System Administrator
- I want to: Enforce AES-256 encryption at rest, TLS 1.3 in transit, and RoleBased Access Control (RBAC)
- So that: Sensitive clinical records and metabolic parameters are protected from unauthorized access.

4) Story 2.1:

 PMDSM-2 /  PMDSM-9

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Story 2.1: Algorithmic Weekly Menu Generation

Done ▾

✓ Done



✦ Improve Story

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▽ **Description**

- As a: System Engine
- I want to: Generate a personalized 7-day meal plan calculated within 5% of the subscriber's calorie goal
- So that: Meal recommendations meet nutritional targets and clinical caloric requirements.

5) Story 2.2:

 PMDSM-2 /  PMDSM-10  1  

Story 2.2: Mandatory Allergen Constraint Validation

Done ▾

✓ Done



✦ Improve Story

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▼ **Description**

- As a: System Engine
- I want to: Execute automated allergen checks against all suggested menu ingredients before publishing a draft
- So that: Any dish containing the subscriber's excluded allergens is filtered out prior to review.

6) Story 2.3:

 PMDSM-2 /  PMDSM-11  1  

Story 2.3: High-Concurrency Plan Generation Performance

Done ▾

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✦ Improve Story

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▼ **Description**

- As a: Platform Architect
- I want to: Optimize menu generation throughput to sustain 2.5 seconds execution under peak load
- So that: The subscription engine handles large batches of automated renewals without latency bottlenecks.

7) Story 3.1:

 PMDSM-3 /  PMDSM-12

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Story 3.1: Clinical Dashboard Menu Review

Done ▾

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⚡ Improve Story

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▼ **Description**

- As a: Dietitian
- I want to: Inspect draft weekly plans alongside patient macro-nutrient breakdowns, diabetic thresholds, and portion limits
- So that: I can verify clinical safety and suitability prior to kitchen preparation.

8) Story 3.2:

 Jira work item  

 PMDSM-3 /  PMDSM-13

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Story 3.2: Clinical Item Override & Real-Time Macro Recalculation

Done ▾

✓ Done



⚡ Improve Story

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▼ **Description**

- As a: Dietitian
- I want to: Override specific meal items with eligible allergen-free replacements and clinical notes
- So that: Unsuitable items (e.g., elevated sodium) can be corrected while automatically recalculating weekly macro totals.

9) Story 3.3:

 PMDSM-3 /  PMDSM-14

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Story 3.3: Meal Plan Sign-off & Audit Logging

Done ▾

✓ Done



✦ Improve Story

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▽ **Description**

- As a: Dietitian
- I want to: Formally sign off on a verified meal plan to transition its state to "Approved"
- So that: Kitchen dispatch is scheduled and audit logs record my review credentials.

10) Story 4.1:

 Jira work item  

 PMDSM-4 /  PMDSM-15

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Story 4.1: Subscription Schedule Modifications

Done ▾

✓ Done



✦ Improve Story

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▽ **Description**

- As a: Subscriber
- I want to: Pause, skip, or reschedule an upcoming daily meal delivery from my account portal
- So that: Deliveries match my personal travel schedules and prevent food wastage.

11) Story 4.2:

☒ Jira work item

 PMDSM-4 /  PMDSM-16

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Story 4.2: 12-Hour Dispatch Cutoff Guard

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✦ Improve Story

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▽ **Description**

- As a: System Engine
- I want to: Enforce a strict 12-hour pre-dispatch deadline for all schedule modifications
- So that: Kitchen preparation and logistical dispatch operations are not disrupted by late cancellations.

12) Story 5.1:

☒ Jira work item

 PMDSM-5 /  PMDSM-17

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Story 5.1: Daily Fulfillment Status Tracking

Done ▾

✓ Done



✦ Improve Story

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▽ **Description**

- As a: Subscriber
- I want to: View the real-time operational status of my daily meal package
- So that: I can anticipate when my food is prepared, dispatched, and delivered.

13) Story 5.2:

☒ Jira work item

⚡ PMDSM-5 / ☒ PMDSM-18

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Story 5.2: Delivery Milestone Event Notifications

Done ▾

✓ Done

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⬆ Improve Story

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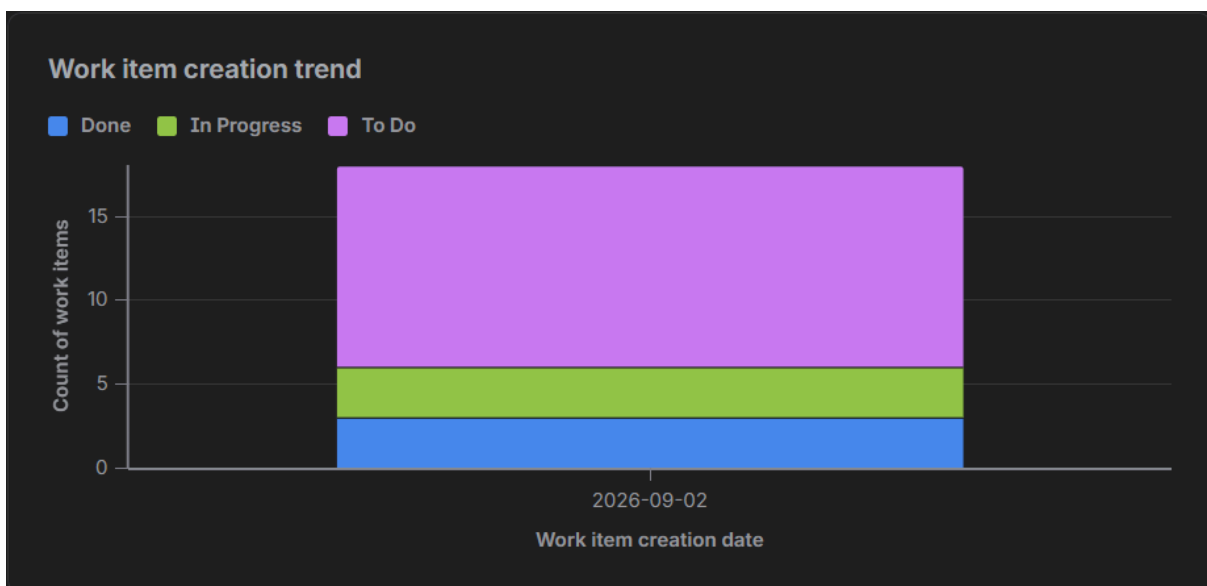
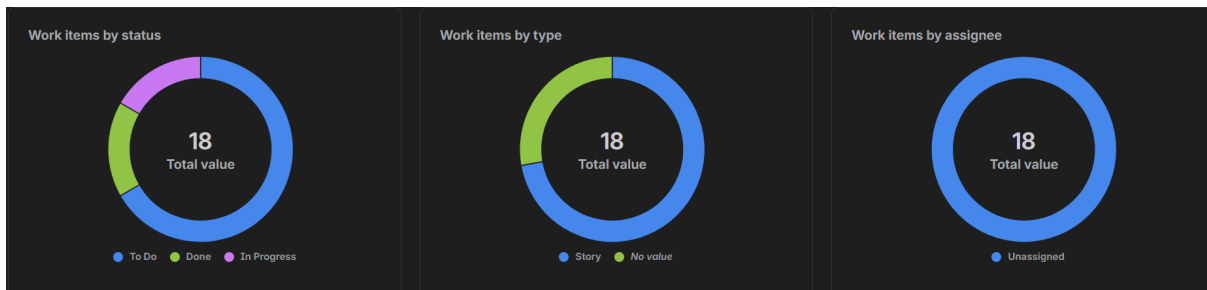
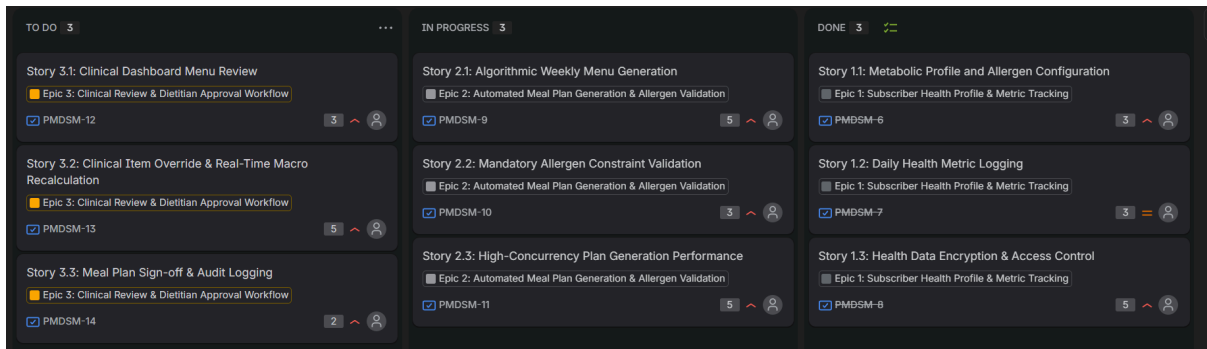
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▽ Description

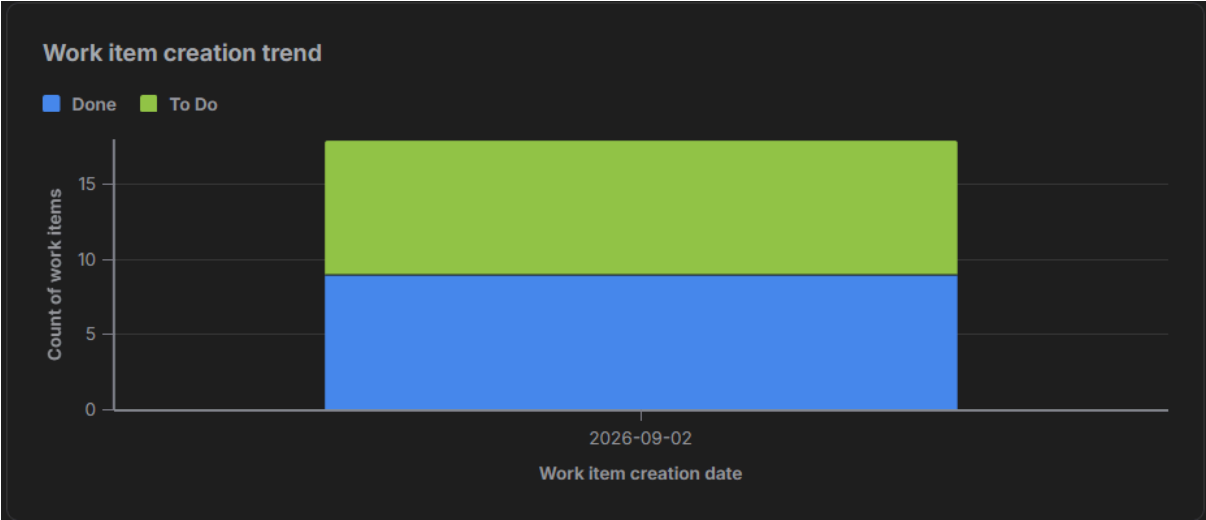
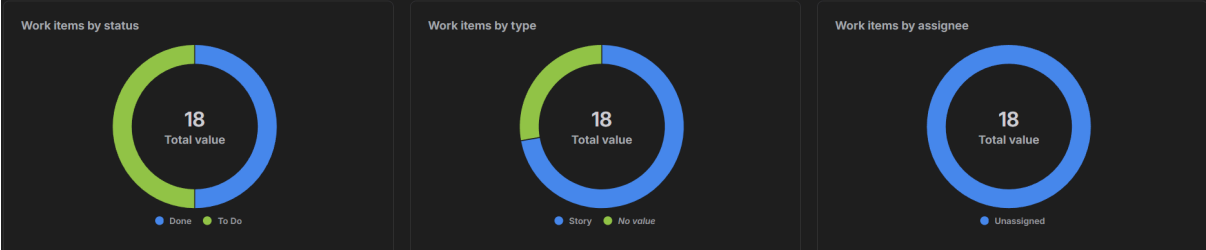
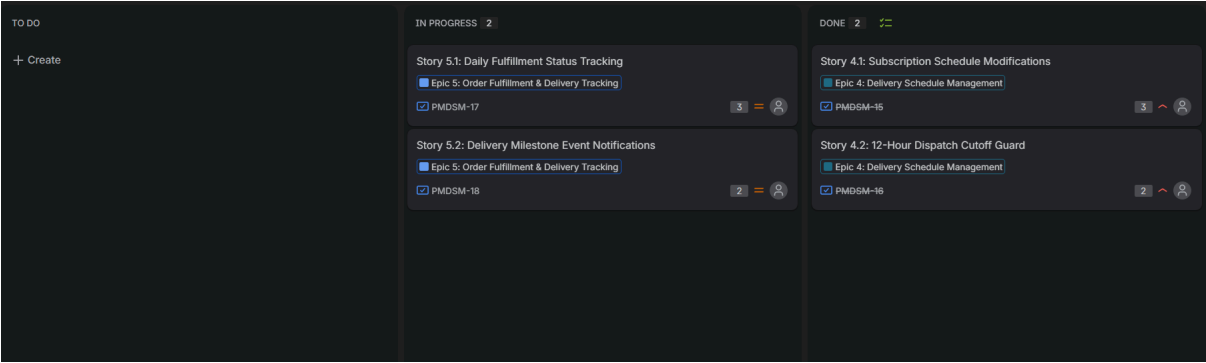
- As a: Subscriber
- I want to: Receive instant push/SMS alerts whenever my meal fulfillment status changes
- So that: I can coordinate receipt of perishable dietary packages promptly.

Sprint, Burnout and Summary:

Week 1:



Week 2:



Comments after sprints:



Rahul Saran Sharma

1 second ago

Sprint completed successfully. The claim submission feature is now fully operational in the user interface.

We have validated that the system correctly verifies the PNR and Baggage Tag Number against the mock airline database, ensuring no invalid submissions are accepted.



Rahul Saran Sharma 

1 second ago

All user stories under this Epic have been successfully completed and moved to Done. The cross-referencing algorithm is now working as expected, accurately matching lost reports with unrouted luggage logs. Furthermore, Baggage Service Agents are able to perform manual status updates without issues, and the system properly records their agent ID along with the timestamp for every update.



Reflection / Conclusion

Transitioning the **Personalized Meal & Diet Subscription Manager** from a problem statement into a structured Jira backlog provided valuable insight into the practical application of Agile methodology. By breaking down the high-level requirements into specific Epics and User Stories, the focus shifted toward the needs of the **subscriber and dietitian**, while clearly defining the value delivered by each feature. The system addresses key requirements such as personalized meal-plan generation, allergen validation, health-metric tracking, and delivery management.

Simulating the sprint cycle demonstrated how development teams prioritize work and manage complexity. Assigning Story Points using the Fibonacci sequence encouraged realistic estimation of effort and uncertainty across tasks, from generating personalized meal plans and validating dietary constraints to managing delivery schedules. Working with the **Active Sprints** board and tracking sprint progress through Jira reports also demonstrated the importance of visual project tracking for maintaining team alignment, monitoring progress, and meeting sprint objectives.

Overall, this lab successfully connected theoretical **Requirements Engineering, UML Use-Case Modelling, and Agile software development concepts** with practical project-management tools. It provided hands-on experience in transforming requirements into actionable backlog items, organizing development work into sprints, and tracking progress throughout the software development process.

Burndown Chart:

